Market Requirements Document (MRD)

CareQueue - Hospital Digital Token and Queue Management System

Field	Value
Document	Market Requirements Document
Product	CareQueue - Hospital Digital Token and Queue Management System
Problem statement	Patients wait in queues for registration and billing. Build a digital token and queue-management system that displays the current token and estimated waiting time.
Version / Status	v1.0 - Draft for hackathon submission
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Team	Future Coders
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How to read this: statements tagged **[A#]** are **assumptions** we have not yet verified with real clinics. They are listed with a validation plan in Section 6. Everything else comes from the problem statement or direct observation.

1. Target User

Primary user: the **walk-in patient** at a small clinic with 1-2 counters, waiting to register or pay a bill.
Secondary user: the **receptionist** who runs registration and billing, often alone.

Clinic profile: private clinic or polyclinic, 1-2 service counters, about 30-150 patients/day **[A1]**, one reception PC or laptop and a TV/monitor available **[A2]**, no IT staff.

2. Personas

Persona	Who	Needs	Today's frustration
Ravi, 34 - patient	Office worker, smartphone user, visits before work	Know how long he'll wait so he can sit or step out	Stands near the counter for 30-40 min with no update
Kamala, 68 - patient	Retired, weak hearing, little phone use	Sit down and be clearly told when it's her turn	Misses her name when called; tiring to stand
Priya, 26 - receptionist	Handles registration and billing alone	Serve patients fast without crowd or arguments	Crowd at the desk, repeated "how long?" questions, "I was first" disputes

3. The Need

3.1 Problem

Patients wait in queues for registration and billing with **no visibility of whose turn it is or how long they will wait**. The result:

- Patients crowd the counter so they don't miss their turn.
- Receptionists lose time answering "how much longer?" **[A3]**.

- Some patients leave before being served **[A4]**.
- Order disputes arise because there is no visible, fair record of who came first.

3.2 Need statement

Walk-in patients need to know the token being served now and roughly how long their own wait will be, so they can wait calmly instead of crowding the counter. Receptionists need a one-click way to call the next patient.

3.3 Why this is a real need

- It is stated directly in the problem statement as an existing hospital pain point.
- Queues at registration and billing are a common, visible experience in Indian clinics and hospitals.
- Larger hospitals already pay for token displays, which shows the need exists; small clinics are left out on cost **[A5]**.

4. Current Alternatives

Alternative	What works	Why it fails small clinics
Paper slips / register / calling names	Free, no setup	No wait estimate, names missed, no fair order, no data
Hardware token machine + display	Clear now-serving display	Upfront hardware cost **[A5]**, no wait estimate, no phone tracking
Full Hospital Information System (HIS)	Complete records and billing	Expensive and complex for 1-2 counters
Generic queue apps (salons, banks)	Remote tracking	Not designed for registration -> billing; often need app install or subscription

4.1 Gap we fill

A **browser-only** system (no extra hardware) that shows the **current token and an estimated waiting time based on real service speed**, and takes under 10 minutes to set up.

5. Market Requirements

ID	Requirement
MR-1	Every patient gets a unique, ordered token for the queue they join (registration or billing).
MR-2	The waiting area clearly shows the token being served now.
MR-3	Each waiting patient can see an estimated waiting time.
MR-4	Runs on hardware the clinic already has, at very low or no cost **[A2]**.

6. Assumptions Register

#	Assumption	How we will validate	Risk if wrong
A1	Target clinics see 30-150 patients/day	Ask 3-5 local clinics for daily counts	Wrong load estimates
A2	Clinics have a PC/laptop and a TV/monitor	Visit / call clinics; check reception setup	Need to bundle a cheap tablet/TV

#	Assumption	How we will validate	Risk if wrong
A3	Wait questions take noticeable receptionist time	Tally sheet at a reception desk for 1 hour	Weaker staff benefit
A4	Some patients walk out due to long, unclear waits	Ask receptionists; short patient survey	Lower business value
A5	Small clinics avoid token machines due to cost	Price check + ask clinic owners	Competitors are cheaper than thought
A6	Patients will scan a QR code to track their token	Observe pilot: share of patients who scan	Phone tracking deprioritised